BioGraph AI

An Explainable Evidence-Ranking System for Drug-Disease Relationship Exploration

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Research-use disclaimer: BioGraph AI organizes public biomedical evidence. It does not diagnose disease, recommend treatment, or establish that a drug is safe or effective for any patient.

Abstract

Biomedical evidence relevant to drug discovery is fragmented across disease-target associations, drug mechanisms, clinical-development records, and scientific literature. I built BioGraph AI to bring those pieces into one inspectable workflow. The system creates disease-target-drug paths from Open Targets data, retrieves publication context from PubMed through NCBI E-utilities, and assigns transparent evidence scores that can be decomposed into biological, literature, and clinical components. Two scoring modes were evaluated: a Full Evidence Model and a reduced Prediction Model that intentionally excludes clinical stage and clinical-report counts. The final benchmark contained 50 known therapeutic drug-disease relationships and 50 screened mismatched controls. Controls were filtered to remove direct clinical evidence, clinical reports, and direct overlap between known drug targets and disease-associated targets. The Full Evidence Model produced mean scores of 91.44 for known pairs and 23.48 for controls; the Prediction Model produced 58.76 and 23.48, respectively. On this benchmark, the two groups were completely separated. The result is best interpreted as strong discrimination on a deliberately screened test set, not as proof of prospective drug-discovery accuracy. The project demonstrates that a small, transparent knowledge-graph system can combine public biomedical sources into an evidence trail that is easy to inspect, challenge, and reproduce.

Keywords: biomedical knowledge graph; drug repurposing; Open Targets; PubMed; explainable scoring; evidence aggregation; computational biology

1. Introduction

A single drug-disease question can require information from several different places. A disease may be linked to dozens of genes or proteins, a drug may act on several targets, and the strongest clinical or literature evidence may use different identifiers. I started BioGraph AI because I wanted one interface that showed not only a possible connection, but also the path that created it.

Knowledge graphs are useful for this problem because they represent entities such as diseases, targets, drugs, and publications as connected nodes rather than isolated rows in a table. Recent work on biomedical knowledge graphs has shown their value for organizing heterogeneous data and for drug-repurposing research, while also emphasizing the need for explainability and careful evaluation [4,5].

The project therefore focuses on traceability rather than a black-box prediction. A user can search for a disease, inspect associated targets, follow a target to a drug or clinical candidate, open the underlying Open Targets record, and review publication evidence. The ranking model is deterministic: every score can be explained in terms of specific components.

The main research question is whether this transparent evidence-ranking approach can separate known therapeutic drug-disease relationships from screened mismatched controls, and whether that separation remains when direct clinical-development information is removed.

2. Background and Related Work

2.1 Public biomedical evidence resources

Open Targets is an open, publicly available platform for systematic target identification, annotation, and prioritization. Its recent platform update emphasizes therapeutic-hypothesis building by integrating evidence across multiple public sources [1]. PubMed and the NCBI Entrez Programming Utilities provide programmatic access to biomedical literature [2]. Drug identifiers and mechanism information exposed

through Open Targets use ChEMBL, a curated database of bioactive molecules and drug-discovery data [3].

2.2 Knowledge graphs in drug repurposing

Drug-repurposing research increasingly uses knowledge graphs to integrate diseases, genes, proteins, drugs, pathways, and publications. A 2024 review of knowledge graphs for drug repurposing highlights both their ability to connect heterogeneous evidence and the importance of explanations that make predictions understandable to researchers [4]. More advanced systems can learn graph representations and perform link prediction across multiple biomedical layers [5]. BioGraph AI is intentionally simpler: it uses a small evidence graph and deterministic scoring as an explainable baseline rather than a learned graph neural network.

3. Development Path and Design Decisions

The final system was not designed in one pass. Several problems only became obvious after I tried to make the first version work with live data. Those iterations shaped the final research design.

Graph first, scoring later: The first usable version only displayed a disease and its associated targets. I added drug nodes only after the graph interaction was stable, because it was easier to debug one evidence layer at a time.

Adapting to the current Open Targets schema: An early drug query used a retired field. The current platform exposes drug and clinical-candidate relationships through drugAndClinicalCandidates, so the backend had to be updated to the current schema before the drug layer would work reliably.

Keeping explanations attached to nodes: The graph became much more useful once clicking a node showed the full name, association score, clinical stage, reason for connection, source link, and publications. This changed the project from a visualization into an evidence-inspection tool.

Separating evidence summary from prediction: The first score used clinical stage directly. That is useful for summarizing how established a relationship is, but it can leak the answer when evaluating known therapeutic pairs. I therefore created a second Prediction Model that removes clinical stage and clinical-report counts.

Fixing the benchmark controls: A pilot benchmark exposed a more serious problem: automatically mismatched controls sometimes turned out to be clinically meaningful relationships. For example, some control pairs had clinical-stage evidence or target overlap. The final benchmark therefore screened controls and replaced pairs with direct clinical evidence or direct target overlap.

These revisions are important because they show why the final benchmark is stricter than the early prototype. The system was changed in response to observed failure modes rather than only to improve the final metric.

4. System Architecture

BioGraph AI is a lightweight web application. The backend is implemented in Python using FastAPI, and the interactive network is rendered in the browser with Cytoscape.js. Open Targets supplies structured biomedical relationships, while NCBI E-utilities are used to retrieve publication titles and abstracts.

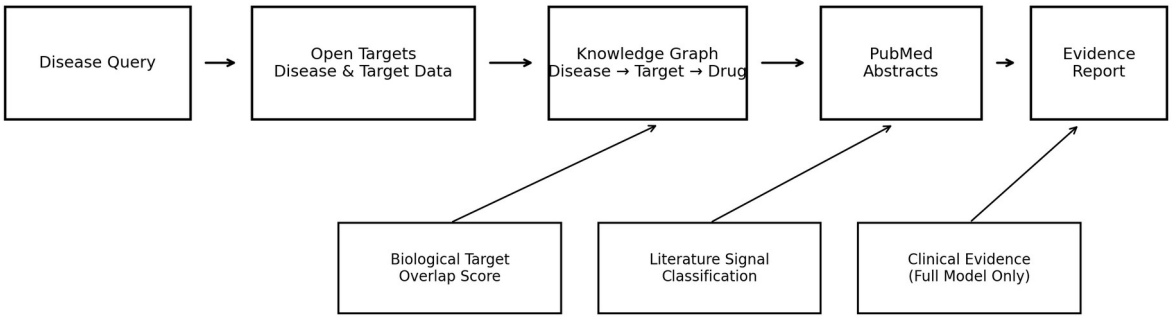


Figure 1. BioGraph AI data flow. A disease query is expanded into target and drug relationships, then enriched with literature and evidence scoring.

4.1 Knowledge graph construction

A disease name is first resolved to an Open Targets disease identifier. The backend requests associated targets and stores each target with its symbol, name, biotype, function description, and disease-association score.

For each displayed target, the system retrieves drugs and clinical candidates. A drug is shown in the disease graph when it appears in the disease clinical data and is also linked to a displayed target. The resulting path is Disease -> Target -> Drug.

Each node keeps its source identifier. This allows the interface to link directly back to Open Targets and to retrieve publication information for the selected entity.

4.2 Evidence panel

Clicking a node opens a side panel containing human-readable details rather than only an identifier. Drug nodes additionally trigger an evidence analysis that reports the score, score components, drug-disease literature mentions, abstract classifications, and source links. The interface is designed so that a user can inspect why a score exists instead of treating it as a probability.

5. Data Sources

Source	Used for	Examples of fields
Open Targets Platform	Disease resolution, target-disease associations, drug/clinical-candidate relationships, mechanisms of action	Disease ID, target ID, association score, clinical stage, clinical reports
PubMed / NCBI E-utilities	Publication retrieval and abstract text	PMID, title, abstract, publication date
ChEMBL identifiers via Open Targets	Drug identity and mechanism links	ChEMBL ID, drug name, molecular targets

Table 1. Main public data sources used by BioGraph AI.

6. Evidence-Ranking Method

The benchmarked ranking core is deterministic. Despite the project name “BioGraph AI,” the evaluated score is not a neural network or large language model. I chose this on purpose so that the first research version would have a baseline that could be audited line by line.

6.1 Biological component

The biological component checks whether known molecular targets of a drug overlap with disease-associated targets. The strongest target-disease association provides the main biological contribution, and additional overlapping targets provide a capped bonus. In the final implementation this component contributes up to 60 points.

6.2 Literature component

Publication volume is logarithmically scaled so that very large literature counts cannot dominate the score. Up to ten retrieved PubMed titles and abstracts are also inspected with a transparent phrase-based classifier. Supportive phrases add evidence; phrases such as “no significant benefit,” “failed to show,” or “lack of efficacy” are treated as possible negative or uncertain signals. Negative phrases are checked first so that expressions such as “no significant improvement” are not incorrectly counted as positive evidence.

6.3 Clinical component

The Full Evidence Model adds clinical-development stage and the number of clinical reports. The Prediction Model removes both features. This creates two different questions: the Full Model asks how much total evidence supports an existing relationship, while the Prediction Model asks whether biological and literature evidence alone still separates known relationships from screened controls.

6.4 Score structure

Prediction score = **literature component** + **biological component** + **abstract-signal component**

Full score = **prediction score** + **clinical-stage component** + **clinical-report component**

All component outputs are clipped so that final scores remain between 0 and 100. The numbers are ranking scores, not calibrated probabilities of treatment success.

7. Benchmark Design

The final benchmark contains 100 pairs: 50 known therapeutic drug-disease relationships and 50 screened mismatched controls. The known set spans oncology, neurology, psychiatry, immunology, respiratory disease, cardiovascular disease, and metabolic disease. Examples include trastuzumab-breast cancer, imatinib-chronic myeloid leukemia, levodopa-Parkinson disease, adalimumab-rheumatoid arthritis, and topiramate-epilepsy.

7.1 Why the controls were screened

The pilot benchmark used automatically rotated disease labels. That looked convenient, but it created mislabeled controls: some supposedly negative pairs had clinical-stage evidence or meaningful target overlap. Rather than keep those pairs and inflate the apparent error rate, I treated that as a benchmark-design failure. In the final version, a candidate control was rejected if Open Targets reported a clinical

stage, any clinical reports, or direct overlap between a known drug target and a disease-associated target.

The final control set should therefore be described as screened mismatched controls, not as experimentally proven negatives. A screened control may still have indirect biology or future therapeutic relevance.

7.2 Evaluation protocol

1. Run the Full Evidence Model and the Prediction Model on the same 100 drug-disease pairs.
2. Record the score for every pair and compare the distributions for known relationships and screened controls.
3. Select the threshold that maximizes F1 score on this benchmark.
4. Report accuracy, precision, recall, F1 score, and pairwise ROC-AUC.
5. Inspect the score margin between the lowest known pair and highest screened control.

Because the threshold was selected on the same dataset used for evaluation, the classification metrics are optimistic and should not be interpreted as external test-set performance. This is a central limitation of the current study.

8. Results

Model	Known mean	Control mean	Lowest known	Highest control	Best threshold	ROC-AUC
Full Evidence	91.44	23.48	68	30	31	1.00
Prediction	58.76	23.48	38	30	31	1.00

Table 2. Final benchmark score separation. Accuracy, precision, recall, and F1 were 1.00 for both models at threshold 31 on this dataset.

The Full Evidence Model produced a mean score of 91.44 for known therapeutic pairs and 23.48 for screened controls. The lowest known Full score was 68, while the highest control score was 30. The reduced Prediction Model produced a mean score of 58.76 for known pairs and 23.48 for controls. Its lowest known score was 38, again above the highest control score of 30. This eight-point minimum margin is more informative than the headline accuracy because it shows that the two groups did not merely separate at a single tied boundary.

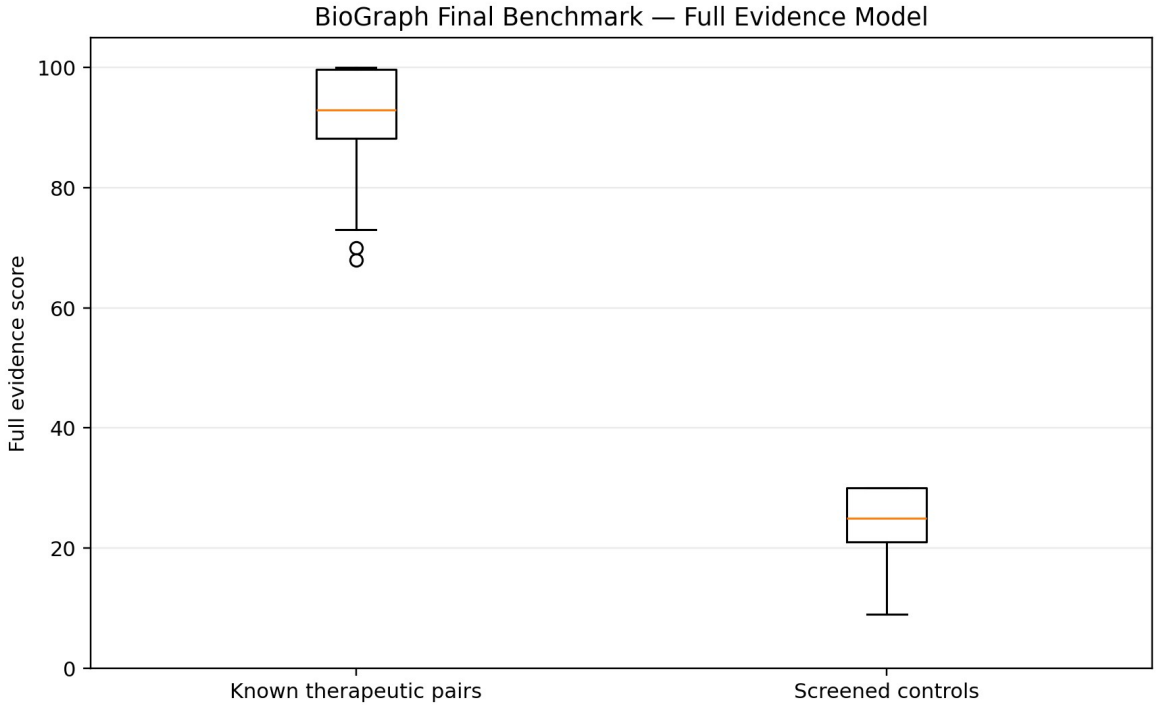


Figure 2. Full Evidence Model score distribution for known therapeutic pairs and screened controls.

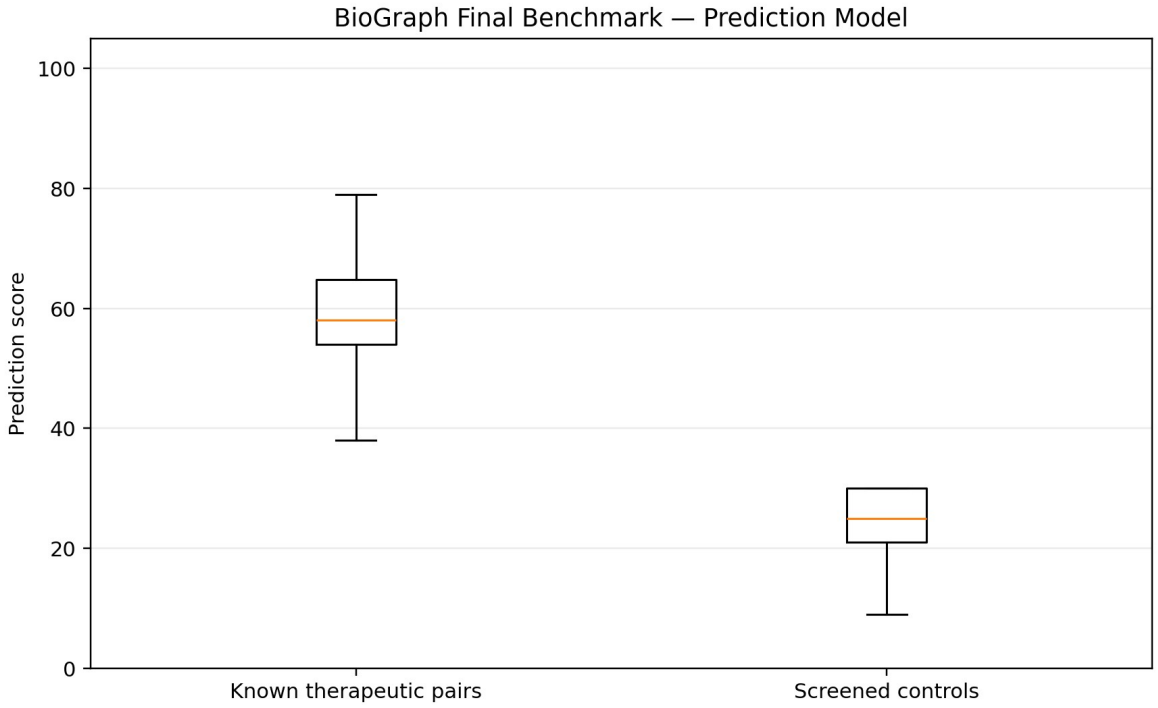


Figure 3. Prediction Model score distribution after removing clinical-stage and clinical-report features.

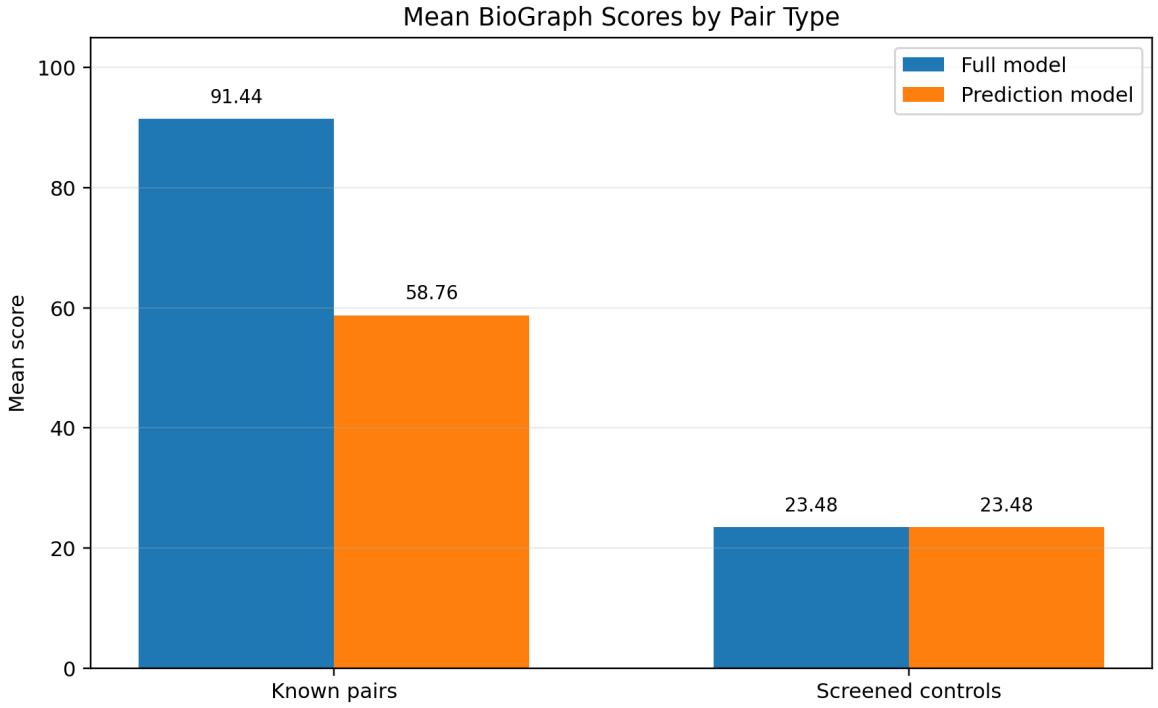


Figure 4. Mean score comparison between known pairs and screened controls.

8.1 Representative pairs

Pair	Group	Prediction score	Key target overlap
lamotrigine - epilepsy	Known	79	SCN1A; SCN2A; SCN8A; SCN3A; SCN9A
topiramate - epilepsy	Known	79	SCN1A; SCN2A; SCN8A; GABRG2; SCN3A
carbamazepine - epilepsy	Known	79	SCN1A; SCN2A; SCN8A; SCN3A; SCN9A
imatinib - chronic myeloid leukemia	Known	77	ABL1; BCR; KIT; PDGFRB
secukinumab - psoriasis	Known	38	IL17A
sildenafil - pulmonary arterial hypertension	Known	39	PDE5A
insulin - type 1 diabetes mellitus	Known	41	INSR
donepezil - melanoma	Screened control	30	None
levodopa - non-small cell lung carcinoma	Screened control	30	None
erlotinib - epilepsy	Screened control	30	None

Table 3. Representative high, low, and boundary cases from the Prediction Model.

9. Discussion

9.1 What the result does show

The final benchmark shows that the selected known therapeutic relationships receive substantially higher evidence scores than the selected screened controls. More importantly, the separation remains when clinical stage and clinical-report counts are removed. That suggests the biological-target and literature components are contributing meaningful discriminatory signal rather than the system simply recognizing approved indications from clinical metadata.

9.2 What the result does not show

A perfect score on this benchmark does not mean BioGraph AI has 100% accuracy at discovering drugs. The controls were screened to remove direct target overlap and clinical evidence, so the final classification task is easier than prospective drug repurposing. The threshold was also selected on the same dataset. A genuinely predictive claim would require an unseen test set, temporal validation, or prospective evaluation.

9.3 Why explainability matters here

The most useful property of the system is not the number 0-100 by itself. A researcher can inspect which target created the path, how strong the disease association is, whether the drug has clinical evidence, which papers were retrieved, and which phrases affected the literature signal. This makes disagreement possible: a user can inspect the evidence and decide that the heuristic classification is wrong.

10. Limitations and Threats to Validity

No independent hold-out set: The decision threshold was optimized on the same benchmark used for reporting performance. This can substantially overestimate generalization.

Screened controls are not true biological negatives: The controls were filtered mismatches, not experimentally proven failed therapies. Some may still have indirect biology or future relevance.

Selection bias in known pairs: The positive set was manually assembled from familiar therapeutic relationships and is not a random sample of all drug indications.

Simple literature classifier: Keyword matching cannot reliably understand study design, dosage, population, endpoint quality, preclinical versus clinical context, or complex negation.

Publication-count ambiguity: A large literature count indicates attention, not efficacy. Highly studied drugs and diseases can produce strong literature signals for unrelated reasons.

Database dependence: Open Targets and PubMed are updated over time. Scores can change as source data and schemas change.

No calibration: A score of 80 should not be interpreted as an 80% probability of success. The score is only a ranking heuristic.

11. Reproducibility

The project was designed so that the benchmark can be rerun from code rather than reconstructed manually. The saved benchmark table contains the drug and disease identifiers, pair label, score components, target-overlap information, publication count, abstract classification counts, clinical stage,

and clinical-report count for every pair. The web application and benchmark use the same Open Targets identifiers and the same scoring functions.

Core software stack: Python, FastAPI, requests, Cytoscape.js, Open Targets GraphQL API, and NCBI E-utilities. The final benchmark CSV should be archived together with the code so that later changes to public databases can be compared against the October 2026 results.

12. Future Work

- Freeze the scoring formula and evaluate it on a completely held-out benchmark that is not used for threshold selection.
- Perform temporal validation: construct scores using evidence available before a historical cutoff, then test whether later clinical successes rank highly.
- Add failed clinical candidates and biologically plausible hard negatives rather than relying mainly on mismatched controls.
- Compare BioGraph with simpler baselines such as literature count only, target overlap only, and clinical evidence only.
- Replace the keyword literature classifier with a biomedical language model, but keep citation-level evidence and compare the new model against the deterministic baseline.
- Evaluate ranking metrics such as precision@k and mean reciprocal rank for candidate lists rather than only binary pair classification.
- Record source timestamps and data-version information so every graph can be reproduced exactly.

13. Responsible Use

Biomedical software can look more authoritative than it is. BioGraph AI therefore presents evidence paths and source links rather than treatment recommendations. The project should be used to generate questions for research, not to make patient-specific decisions. Any future version that uses a learned model or large language model should preserve the same principle: evidence should remain traceable to sources, uncertainty should be explicit, and the system should be evaluated before stronger claims are made.

14. Conclusion

BioGraph AI began as a disease-target visualization and developed into an explainable evidence-ranking system after several rounds of debugging, schema changes, and benchmark redesign. The final system links diseases to biological targets and drugs, retrieves publication context, and produces transparent evidence scores. In the final 100-pair benchmark, known therapeutic relationships were completely separated from screened controls in both the Full Evidence Model and the reduced Prediction Model. The reduced model is the more important result because it removes direct clinical-stage information. Even so, the current benchmark is an internal evaluation, not proof of prospective drug-repurposing performance. The strongest next step is independent validation on unseen and more difficult examples. The main contribution of the project is therefore not a claim of “drug discovery,” but a reproducible and inspectable framework for organizing biomedical evidence into hypotheses that can be questioned and tested.

References

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Appendix A. Final Benchmark Summary

Final dataset size: 100 pairs (50 known therapeutic relationships and 50 screened mismatched controls).

Final control-screening rule: reject a candidate control if it has a non-empty clinical stage, a positive clinical-report count, or direct overlap between a known drug target and a disease-associated target in the queried Open Targets data.

Full Evidence Model: known mean = 91.44; control mean = 23.48; lowest known = 68; highest control = 30.

Prediction Model: known mean = 58.76; control mean = 23.48; lowest known = 38; highest control = 30.

At threshold 31 on this benchmark, both models produced 50 true positives, 50 true negatives, 0 false positives, and 0 false negatives. Because threshold selection and evaluation used the same benchmark, these classification metrics should be treated as descriptive rather than as external validation.

Appendix B. Author Note

This paper documents the actual development path of the BioGraph AI project, including unsuccessful queries, changes to the Open Targets schema used by the application, and redesign of the benchmark after problematic controls were discovered. These details are included because they are part of the research process and because they make the final result easier to audit. Any submission of this paper should follow the receiving school, competition, journal, or platform rules regarding disclosure of software, writing, or AI-assisted tools used during development.